

REVIEW

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Machine learning in cancer imaging for enhanced precision in diagnosis and therapy

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Abstract

Machine learning (ML) is transforming cancer imaging by enhancing diagnosis, automating image analysis, and improving treatment planning. This review explores key ML applications, including tumor detection, radiomics, multi-modal imaging, and therapy monitoring. We discuss fundamental ML techniques, deep learning architectures, and data preprocessing strategies essential for medical imaging. ML-driven approaches have improved tumor segmentation, feature extraction, and computer-aided diagnosis across various cancer types. In cancer therapy, artificial intelligence (AI) aids radiotherapy planning, treatment response prediction, and real-time image-guided interventions. However, challenges such as data scarcity, model bias, and regulatory hurdles limit clinical adoption. Emerging solutions include explainable AI, federated learning for data privacy, and quantum computing for advanced imaging. Addressing these challenges through interdisciplinary collaboration will accelerate AI integration into clinical practice, enhancing cancer diagnosis and treatment.

Keywords Machine learning, Cancer imaging, Deep learning, Radiomics, Artificial intelligence, Diagnosis, Therapy planning, Medical imaging

1 Introduction

Medical imaging is an essential component of oncology, providing critical information for cancer diagnosis, staging, treatment planning, and therapy monitoring. Various imaging modalities, including computed tomography (CT), magnetic resonance imaging (MRI), positron emission tomography (PET), ultrasound (US), and histopathology, are routinely used in clinical practice to assess tumor characteristics and guide therapeutic decisions [1–3]. The accuracy and reliability of imaging interpretation are crucial for effective patient management; yet conventional image analysis relies heavily on human expertise, which can be subject to inter-observer variability and time constraints [4].

Machine learning (ML) has gained significant attention in medical imaging due to its ability to automate and enhance image analysis [5, 6]. ML methods, particularly deep learning (DL) approaches such as convolutional neural networks (CNNs), have



demonstrated promising results in tasks such as tumor detection, segmentation, classification, and outcome prediction [7, 8]. These techniques have the potential to improve diagnostic accuracy, reduce workload for radiologists and radiation oncologist, and facilitate personalized treatment planning. Moreover, ML-driven radiomics [9, 10], which involves extracting quantitative features from medical images, has opened new possibilities for non-invasive tumor characterization and treatment response assessment.

In cancer therapy, ML applications extend beyond diagnosis to treatment planning and monitoring. Artificial intelligence (AI)-assisted radiotherapy planning [11], predictive modeling of treatment responses [12], and real-time image-guided interventions [13] are actively being explored to enhance precision and adaptability in oncology. While ML-based models show potential, their clinical translation is challenged by factors such as data availability, model interpretability, and regulatory approval [14]. Addressing these challenges is essential for integrating ML into routine clinical workflows and ensuring its reliability in medical decision-making.

This topical review provides an overview of ML applications in cancer imaging, discussing key methodologies, current advancements, and challenges. The role of ML in tumor detection, segmentation, radiomics, and therapy planning is examined, along with emerging trends such as explainable AI (XAI) [15], federated learning [16], and the integration of quantum computing [17]. By summarizing the current state of research and identifying future directions, this review aims to contribute to the ongoing development of ML-driven cancer imaging solutions.

While prior reviews have offered valuable overviews of ML in cancer diagnostics and treatment [18–20], our review distinguishes itself by focusing on the integration of ML architectures into closed-loop, adaptive imaging-treatment workflows. We also highlight translational bottlenecks, such as data privacy, explainability, and workflow feedback cycles, as key enablers for clinical adoption, areas often not emphasized in more algorithm-centric or diagnostic-focused reviews.

This review makes several scientific contributions that extend beyond architectural surveys and aim to bridge technical development with clinical translation in cancer imaging and therapy. First, it proposes a workflow-oriented synthesis that maps ML models to distinct stages of the oncology pipeline, from image acquisition and preprocessing to radiogenomic modeling and therapy monitoring. This structure provides conceptual clarity and emphasizes clinical integration. Second, the review identifies key translational challenges, such as interpretability, generalizability, and feedback-driven adaptation, and integrates them into the evaluation of current ML methods. Third, it contextualizes emerging fields such as XAI, federated learning, and quantum machine learning within the oncology domain, offering insights into how these technologies may enhance robustness, privacy, and computational efficiency. Finally, the review highlights the critical need for closed-loop, adaptive AI systems that respond dynamically to imaging feedback, offering a vision for future intelligent platforms that not only predict outcomes but iteratively refine their guidance based on treatment response. These contributions collectively aim to support interdisciplinary advancement by aligning technical innovation with clinical priorities and regulatory considerations.

Recent reviews have provided valuable summaries of deep learning architectures and diagnostic performance benchmarks in cancer imaging, with particular emphasis on tumor detection, segmentation, radiomics feature extraction, and multi-modal image

analysis [21–24]. Many of these surveys focus on algorithmic innovation and retrospective performance metrics, offering detailed comparisons of model architecture and imaging modalities. However, diagnosis, treatment planning, and therapy monitoring are often discussed as largely independent components, and comparatively less attention is given to how ML models function across integrated, end-to-end clinical workflows. Furthermore, challenges related to real-world deployment such as interoperability with radiotherapy planning systems, regulatory approval pathways, and constraints imposed by real-time clinical environments are frequently acknowledged only at a high level rather than examined in depth.

In contrast, the present review adopts a workflow-oriented and translational perspective that maps ML methods across the full oncology imaging–treatment continuum, from image acquisition and tumor characterization to therapy planning, treatment monitoring, and adaptive intervention. Rather than organizing content solely by algorithm type or imaging modality, this review emphasizes how ML models interact within closed-loop, feedback-driven clinical workflows that evolve over the course of treatment.

A further distinguishing feature of this review is its explicit focus on clinical translation. Key barriers to routine deployment, including workflow integration, model interpretability and trust, regulatory and liability considerations, and post-deployment monitoring, are examined in detail alongside methodological advances. Emerging paradigms such as explainable AI, federated learning, and quantum-enhanced machine learning are discussed not only as technical innovations, but also in terms of their potential to address practical deployment challenges [25–27]. By integrating methodological progress with system-level design and clinical constraints, this review complements existing literature and provides a unified framework for understanding how machine learning can transition from promising research tools to reliable components of routine cancer care.

1.1 Literature search strategy and review methodology

This review was conducted as a structured narrative review with systematic elements to ensure comprehensive coverage of recent advances in ML-based cancer imaging and therapy. The literature search focused on peer-reviewed journal articles and high-impact conference proceedings published primarily between 2019 and 2026, with particular emphasis on studies from 2024 to 2026 to capture the latest developments.

Searches were performed using major scientific databases including PubMed, Web of Science, IEEE Xplore, and Google Scholar. Key search terms and combinations included *machine learning*, *deep learning*, *artificial intelligence*, *cancer imaging*, *radiomics*, *medical image segmentation*, *radiotherapy planning*, *treatment response prediction*, *explainable AI*, and *federated learning*. Reference lists of relevant review articles were also screened to identify additional pertinent studies.

Studies were selected based on their relevance to cancer imaging and therapy, methodological rigor, clinical relevance, and contribution to understanding translational challenges. Priority was given to high-quality studies published in leading journals and conferences in medical imaging, oncology, and artificial intelligence. Articles focusing exclusively on non-oncologic imaging or lacking relevance to clinical workflows were excluded.

Although this review does not follow a full PRISMA systematic review protocol or quantitative meta-analysis, the search and selection process was guided by PRISMA principles of transparency, reproducibility, and clarity [28]. The goal was to provide a balanced and critical synthesis of the literature while emphasizing emerging trends, clinical translation barriers, and future research directions beyond algorithmic performance alone.

2 Fundamentals of machine learning in medical imaging

ML has been increasingly applied to medical imaging, offering methods to automate image analysis, enhance diagnostic accuracy, and optimize treatment planning. ML techniques are broadly categorized into supervised, unsupervised, and reinforcement learning, each serving different purposes in image-based medical applications [29]. Moreover, DL architectures such as CNNs, recurrent neural networks (RNNs) [30], and generative adversarial networks (GANs) [31] have demonstrated significant potential in medical image processing. However, the success of these models depends on effective data acquisition, preprocessing, and annotation.

2.1 Basic concepts of machine learning in imaging

ML involves training algorithms to recognize patterns in medical images and make predictions based on these patterns [32]. In cancer imaging, ML models learn from labeled datasets to classify tumors, segment anatomical structures, and predict patient outcomes. The primary ML paradigms include supervised learning, unsupervised learning, and reinforcement learning, each with distinct applications in medical imaging [33].

Supervised learning is widely used in medical imaging for tasks such as tumor classification and segmentation. In this approach, models learn from labeled datasets where the correct output (e.g., presence or absence of a tumor) is known [34]. Classification models, such as support vector machines (SVMs) and DL-based CNNs, are commonly used for diagnosing cancer from radiological images [35, 36]. Segmentation models, such as U-Net and Mask R-CNN, help delineate tumors and other anatomical structures, assisting radiologists in treatment planning [37, 38]. For classification tasks such as tumor detection, deep networks are typically optimized using cross-entropy loss [39]:

$$L_{CE} = - \sum_i y_i \log(\hat{y}_i) \quad (1).$$

In Eq. 1, L_{CE} is the number of classes, y_i is the ground truth label (one-hot encoded), and $\hat{y}_i$ is the predicted softmax probability. In weakly supervised learning, labels are available at the image level, but not for individual regions or pixels. One common approach is to model the probability of the whole image being positive as:

$$P(y = 1 | X) = 1 - \prod (1 - p_i) \quad (2)$$

In Eq. 2, p_i is the model's predicted probability for instance i , and X is the set of image patches. This formulation assumes that at least one instance in a positive bag is positive.

Unsupervised learning is applied when labeled data is scarce. This approach identifies patterns in imaging data without predefined labels [40]. Clustering techniques, such as k-means and hierarchical clustering, can group similar lesions or tissue types, aiding in tumor characterization [41]. Anomaly detection algorithms, including auto-encoders

and one-class SVMs, are used for identifying rare tumor presentations or early disease manifestations that may not be well-represented in training datasets [42].

Reinforcement learning (RL) is gaining interest in adaptive medical imaging and treatment planning. In RL, an agent learns through trial and error by interacting with an environment and receiving feedback. Applications include optimizing imaging protocols to reduce radiation dose while maintaining image quality, and dynamic radiotherapy treatment planning, where RL can adjust beam configurations based on tumor response over time [43]. Figure 1 provides a graphical comparison of the three primary ML paradigms, namely, supervised, unsupervised, and RL, highlighting their input requirements, learning objectives, and typical applications in cancer imaging. This visual summary complements the textual explanation by clarifying key conceptual distinctions for readers.

2.2 DL architectures used in medical imaging

DL, a subset of ML, has transformed medical imaging with its ability to learn hierarchical features directly from raw images. Several DL architectures have been applied in cancer imaging, each tailored to specific tasks.

CNNs are the backbone of many medical imaging applications due to their ability to automatically extract spatial features from images. They consist of convolutional layers that capture edges, textures, and complex structures, making them highly effective for tasks such as tumor classification and organ segmentation. CNN architectures such as ResNet, VGG, and DenseNet have been used in cancer detection and radiomics feature extraction [44].

While CNNs are effective for spatial analysis, RNNs and Transformers are useful for analyzing sequential medical imaging data, such as dynamic contrast-enhanced MRI and longitudinal scans. RNNs, particularly Long Short-Term Memory (LSTM) networks, can

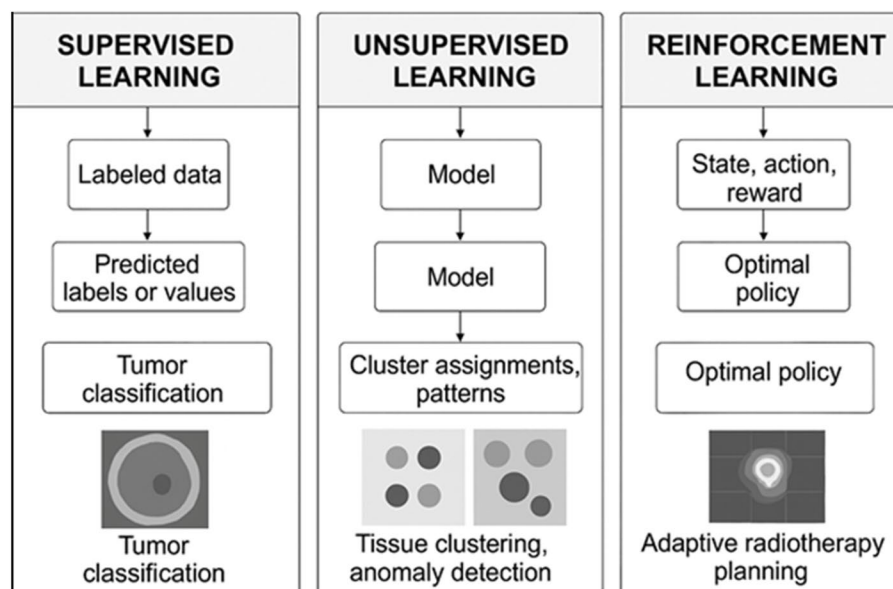


Fig. 1 Conceptual comparison of primary ML paradigms in medical imaging. This diagram illustrates the three main categories of ML: supervised learning, unsupervised learning, and RL, highlighting their distinct input types, learning objectives, outputs, and representative applications in cancer imaging. Supervised learning uses labeled data to predict clinical outcomes such as tumor classification. Unsupervised learning identifies patterns or clusters in unlabeled data, useful for tissue segmentation or anomaly detection. RL learns optimal policies through feedback from actions and is applied in adaptive radiotherapy planning and imaging protocol optimization

model temporal dependencies in imaging sequences [45]. More recently, Transformer-based models, including Vision Transformers (ViTs), have shown promise in capturing long-range dependencies in medical images, improving tumor detection and segmentation accuracy [46].

GANs have been employed in medical imaging for tasks such as image synthesis, denoising, and resolution enhancement. They consist of a generator network that creates synthetic images and a discriminator network that distinguishes real from fake images. GANs have been used to augment training datasets, improve image quality in low-dose CT scans, and generate synthetic MRI images to balance underrepresented tumor types in datasets [47].

2.3 Data acquisition and preprocessing

The performance of ML models in medical imaging heavily depends on data quality and preprocessing techniques. Proper data handling ensures that models generalize well across different patient populations and imaging devices.

Medical images often vary in intensity, resolution, and noise levels due to differences in imaging protocols and equipment. Image normalization techniques, such as histogram equalization and intensity standardization, help reduce variability [48]. Data augmentation methods, including rotation, flipping, and contrast adjustments, are commonly used to artificially increase dataset size and improve model robustness. Denoising techniques, such as wavelet filtering and DL-based denoising autoencoders, help remove artifacts and improve image clarity [49].

Medical imaging datasets are often imbalanced, with certain cancer types or tumor subtypes being underrepresented. Class imbalance can lead to biased ML models that favor the majority class. Techniques such as synthetic minority over-sampling (SMOTE) [50], cost-sensitive learning, and transfer learning help mitigate this issue. Moreover, few-shot learning approaches, including meta-learning and self-supervised learning, are being explored to train models with limited labeled data.

High-quality annotated datasets are essential for training reliable ML models. Image annotations, such as tumor boundaries or classification labels, require expert radiologists, making the annotation process time-consuming and resource-intensive. Crowdsourcing, semi-supervised learning, and active learning strategies are being explored to reduce annotation costs while maintaining accuracy [51]. Collaborative efforts, such as The Cancer Imaging Archive (TCIA) and other publicly available medical imaging datasets, play a crucial role in advancing ML research in oncology [52]. Figure 2 shows a block diagram summarizing the fundamentals of ML in medical imaging. The diagram categorizes ML techniques with color-coded annotations to illustrate the typical application of each model across different cancer types.

Numerous publicly available datasets have played a critical role in advancing ML applications in cancer imaging. These datasets span various imaging modalities and support tasks such as tumor segmentation, lesion classification, and radiogenomics modeling. Table 1 summarizes representative datasets, outlining their imaging types, cancer focus, available annotations, and typical use cases.

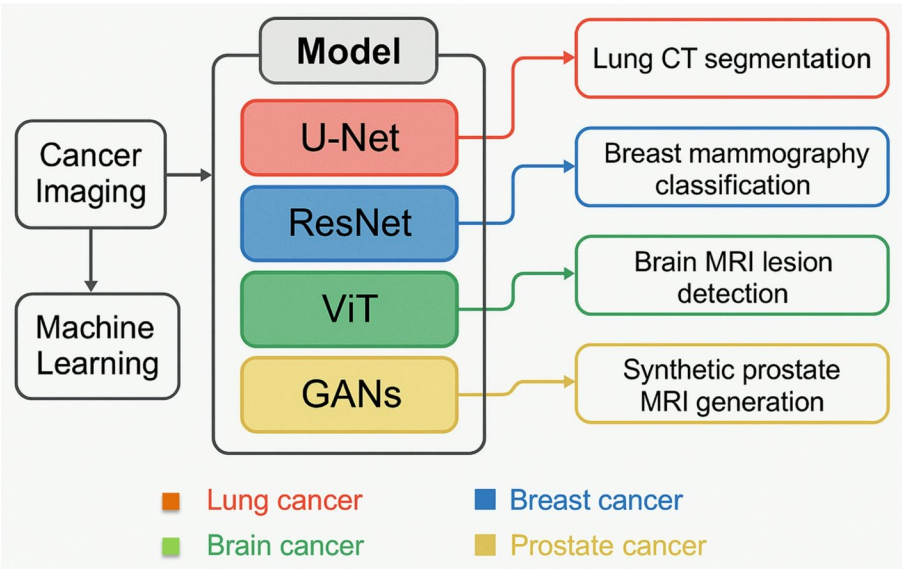


Fig. 2 Color-coded block diagram illustrating machine learning (ML) models and their primary applications in cancer imaging. The diagram maps common ML architectures, U-Net, ResNet, Vision Transformers (ViT), and Generative Adversarial Networks (GANs), to specific cancer imaging tasks using a color-coded scheme. Red, blue, green, and yellow boxes correspond to lung, breast, brain, and prostate cancers, respectively. Call-out boxes specify key use cases: U-Net for lung CT segmentation, ResNet for breast mammography classification, ViT for brain MRI lesion detection, and GANs for synthetic prostate MRI generation. This visualization clarifies the relationship between model types and cancer-specific imaging applications, addressing interpretability and contextual alignment

Table 1 Publicly available datasets commonly used for machine learning in cancer imaging

Dataset name	Modality	Cancer type	Annotation type	Common use case	Source/link
TCGA-GBMLGG	MRI	Glioma (brain)	Tumor masks, grades	Segmentation, grading	The Cancer Imaging Archive
LIDC-IDRI	CT	Lung cancer	Nodule boundaries, malignancy score	Detection, classification	TCIA
BRATS	MRI	Brain tumors	Pixel-wise tumor labels	Multiclass segmentation	MICCAI Challenge
INbreast	Mammography	Breast cancer	Masses, calcifications	Lesion classification	INESC Porto
ProstateX	mpMRI	Prostate cancer	Lesions+ biopsy data	Lesion detection, grading	SPIE-AAPM-NCI
CAMELYON16/17	Histopathology	Breast metastases	WSIs + pixel masks	Metastasis classification	Grand Challenge
NSCLC Radiogenomics	CT	Lung (NSCLC)	EGFR status + images	Radiogenomics modeling	TCIA
Head-Neck-PET-CT	PET/CT	Head and neck cancers	Contours, clinical outcomes	Radiotherapy planning	TCIA

This table summarizes benchmark datasets used in medical imaging applications such as tumor segmentation, lesion classification, radiogenomics, and image preprocessing. The datasets span multiple modalities (e.g., MRI, CT, PET, histopathology) and cancer types, with corresponding annotation formats and use cases. These resources are critical for training, validating, and benchmarking AI models in oncology research

2.4 Evaluation metrics and model validation strategies

Evaluating the performance of ML models in cancer imaging requires the use of well-defined metrics tailored to the specific task at hand, whether classification, segmentation, or regression. For classification tasks such as tumor detection or malignancy prediction, commonly used metrics include accuracy, sensitivity (recall), specificity,

precision, and the F1-score [53], which balances the trade-off between precision and recall. Among these, the area under the receiver operating characteristic curve (AUC-ROC) is often preferred in clinical research because it measures a model's ability to distinguish between classes across different threshold settings, providing insight into overall discriminatory power [54].

For image segmentation tasks, such as delineating tumor boundaries on MRI or CT scans, spatial overlap metrics are essential. The Dice Similarity Coefficient (DSC) is widely used to quantify the overlap between predicted and ground-truth masks, with higher values indicating better agreement. The DSC is used both as a performance metric and loss function in medical image segmentation [55]:

$$DSC = 2 \frac{|P \cap G|}{|P| + |G|} = 2 \frac{\sum_i p_i g_i}{\sum_i p_i + \sum_i g_i} \quad (3)$$

In Eq. 3, P is the predicted binary segmentation mask, G is the ground truth binary segmentation mask, p_i and g_i are the predicted and ground truth voxel labels, respectively.

Its corresponding loss function, the Dice loss (L_{Dice}), is defined as:

$$L_{Dice} = 1 - DSC \quad (4)$$

In metric learning, triplet loss encourages embeddings of the same class to be closer than those of different classes [56]. The loss is formulated as:

$$L_{triplet} = \max(d(a, p) - d(a, n) + \alpha, 0) \quad (5)$$

In Eq. 5, a is the anchor sample, p is a positive (same class), n is a negative (different class), $d()$ is a distance metric (e.g., Euclidean), and α is a margin.

Similarly, the Intersection over Union (IoU) provides a geometric measure of segmentation accuracy. These metrics are critical in radiotherapy planning and surgical navigation, where spatial precision is vital [57].

In regression tasks such as predicting treatment outcomes or tumor volume changes, performance is typically evaluated using mean absolute error (MAE), root mean squared error (RMSE), or R-squared values, depending on the specific application [58]. These metrics assess how closely the predicted continuous outputs align with true clinical values, such as survival duration or dose distribution.

Beyond performance metrics, validation strategies play a crucial role in assessing the generalizability and clinical reliability of ML models. Basic internal validation is often conducted using k-fold cross-validation [59], which splits the training data into multiple folds to assess model consistency. Holdout validation, where a separate portion of the dataset is reserved for final testing, is also common. However, these methods may still suffer from institutional bias. Therefore, external validation, using independent datasets from different institutions or imaging platforms, is considered essential for evaluating robustness across varied clinical settings.

Moreover, prospective validation through clinical trials or pilot implementations provides the most rigorous test of clinical utility [60]. Such evaluations help determine how ML models perform under real-world conditions, including integration with workflows, operator variability, and patient heterogeneity. Additional factors like model calibration, which ensures that predicted probabilities reflect true risk, and uncertainty estimation,

which identifies cases where model confidence is low, are increasingly emphasized in medical AI to support risk-aware decision-making.

Together, the use of appropriate evaluation metrics and rigorous validation strategies forms the foundation for developing clinically trustworthy ML systems. These practices not only enhance reproducibility and transparency but also build confidence among practitioners and regulatory bodies, facilitating the path from algorithm development to bedside deployment.

3 Machine learning applications in cancer diagnosis

ML has transformed cancer diagnosis by enhancing the accuracy and efficiency of image analysis. ML algorithms assist in detecting tumors, segmenting anatomical structures, extracting imaging features for predictive modeling, and integrating multi-modal imaging data. Computer-aided diagnosis (CAD) systems further support radiologists by automating routine tasks and improving diagnostic consistency [61].

3.1 Tumor detection and segmentation

Accurate tumor detection and segmentation are critical for early diagnosis and treatment planning. ML-based techniques automate these processes, improving precision and reducing variability across different radiologists and imaging protocols. CNNs have been widely used for tumor detection due to their ability to learn hierarchical image features. Models such as ResNet, VGG, and EfficientNet have demonstrated high accuracy in detecting lung nodules, breast lesions, and brain tumors [62]. More recently, Transformer-based models, such as Vision Transformers (ViTs) and Swin Transformers, have been explored for lesion detection, leveraging self-attention mechanisms to capture long-range dependencies in imaging data [63].

Segmentation is crucial for outlining tumor boundaries, which impacts both diagnosis and treatment planning. U-Net, a widely used DL architecture, has been successfully applied to segment tumors in MRI, CT, and PET images. Variants such as Attention U-Net and nnU-Net enhance segmentation accuracy by incorporating attention mechanisms, allowing the model to focus on relevant image regions [64, 65]. These models are particularly useful in radiotherapy planning, where precise delineation of tumor margins is required.

ML-based tumor detection and segmentation techniques have been utilized across various cancer types. For lung cancer, CNNs and hybrid DL models are employed to detect pulmonary nodules in CT scans and assess tumor progression [66]. In breast cancer, DL models aid in identifying microcalcifications and masses in mammography and MRI [67]. AI-based segmentation techniques are used to classify gliomas, meningiomas, and metastases from MRI scans in brain tumors [68]. For prostate cancer, ML models analyze multi-parametric MRI (mpMRI) to detect prostate lesions and guide biopsy decisions [69]. Figure 3 shows the workflow of a hybrid DL model for brain tumour classification. The model was applied to the MRI dataset for meningioma, glioma and pituitary tumour [68].

DL architectures, particularly CNNs and Transformers, are preferred for tumor detection and segmentation due to their ability to automatically learn hierarchical and spatial features directly from raw imaging data. CNNs excel at capturing local patterns such as edges and textures, while Transformers leverage self-attention mechanisms to model

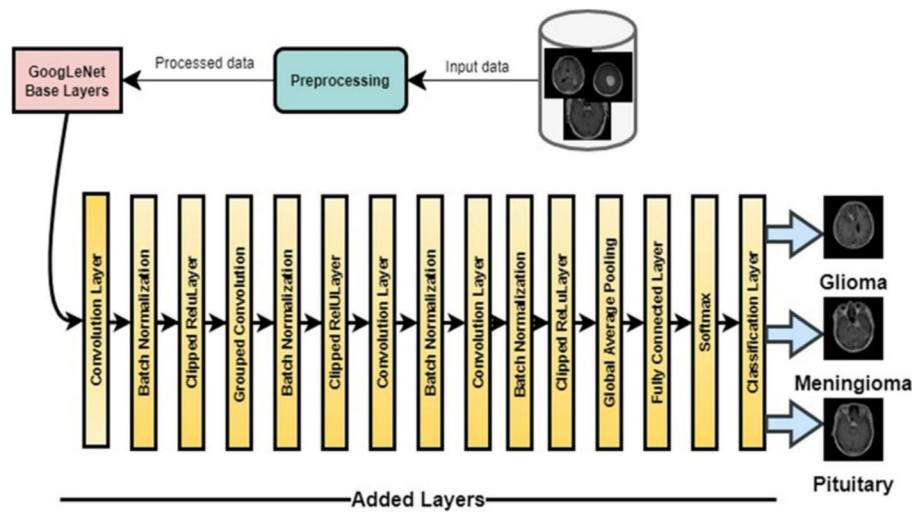


Fig. 3 Deep convolutional architecture for brain tumor classification using transfer learning. The diagram shows a modified GoogLeNet-based model architecture designed for classifying MRI brain scans into glioma, meningioma, and pituitary tumor types. Additional convolutional layers and normalization units are added to improve feature extraction on top of the base model. This architecture exemplifies how transfer learning, coupled with targeted architectural modifications, can adapt pretrained networks to domain-specific cancer imaging tasks, even with limited datasets. <https://www.mdpi.com/1099-4300/24/6/799>. Reproduced from reference [68] under the Creative Commons Attribution 4.0 International License (<https://creativecommons.org/licenses/by/4.0/> (accessed on 12 March 2026)) with no change of the original content

long-range dependencies, which is especially beneficial in complex anatomical contexts. Compared to traditional models like SVMs and random forests, DL models generally achieve higher accuracy in segmentation tasks and scale more efficiently with large, high-dimensional datasets. While SVMs and random forests require manual feature engineering and struggle with pixel-wise prediction, CNNs and Transformers integrate feature extraction and classification in an end-to-end framework, reducing preprocessing overhead and improving generalization. Moreover, advances in graphics processing unit acceleration and parallel computing have made deep learning models more computationally efficient during inference, enabling real-time clinical applications.

3.2 Radiomics and feature extraction

Radiomics refers to the process of extracting a large number of quantitative features from medical images to uncover tumor characteristics that may not be visible to the human eye. ML models analyze these high-dimensional features to predict tumor biology, prognosis, and response to therapy. Radiomics workflows involve extracting shape, texture, intensity, and wavelet-based features from medical images [70]. These features can be used for tumor classification, grading, and treatment response assessment. Feature selection techniques, such as recursive feature elimination (RFE) and principal component analysis (PCA), help reduce dimensionality and improve model interpretability [71]. PCA, a dimensionality reduction technique, transforms radiomic features into orthogonal components that capture the most variance, making it particularly effective in modalities like MRI and PET where high-dimensional texture and intensity features are prevalent. In contrast, RFE iteratively removes less informative features based on model performance, offering greater interpretability and often yielding superior

results in CT-based radiomics where structural features dominate. However, integrating radiomics features into multi-modal models presents unique challenges, including feature heterogeneity, scale mismatches, and modality-specific noise. Harmonizing features across modalities requires careful preprocessing, standardization, and alignment strategies to ensure that combined models do not amplify modality-specific biases. Advanced fusion techniques and cross-modality feature selection frameworks are being explored to address these challenges, enabling more robust and generalizable multi-modal radiomics models.

ML classifiers such as random forests, SVMs, and deep neural networks (DNNs) are used to distinguish between benign and malignant tumors. For example, ML models trained on radiomics features have been employed to predict the aggressiveness of prostate cancer based on mpMRI and to classify breast lesions on mammograms [72].

Radiogenomics combines radiomic features with genomic data to uncover associations between imaging phenotypes and underlying molecular characteristics. As illustrated in Fig. 4, a representative case study demonstrates how medical imaging can be leveraged to predict tumor-driving gene mutations. In this study, a conditional Generative Adversarial Network (cGAN) was employed to generate synthetic MRI data and predict the mutation status of key driver genes in breast cancer patients based on radiogenomic analysis of MRI features [73].

3.3 Multi-modal imaging analysis

Combining data from multiple imaging modalities enhances diagnostic accuracy and provides a comprehensive view of tumor characteristics. DL models facilitate the fusion of different imaging data to improve cancer detection and staging. Different imaging modalities provide complementary information about tumors. For instance, PET-CT combines metabolic and anatomical data [74], while MRI and histopathology jointly improve tumor characterization. AI models integrate these data sources to improve diagnostic precision and reduce ambiguity in complex cases [75]. Combining PET, MRI, and CT data in multi-modal imaging analysis enhances cancer diagnosis by leveraging complementary information from each modality, structural detail from CT, soft-tissue contrast from MRI, and metabolic activity from PET. This integration improves lesion characterization, staging accuracy, and treatment planning compared to single-modality imaging. However, aligning and integrating heterogeneous data introduces significant challenges. Variations in spatial resolution, imaging protocols, and patient positioning can lead to misregistration and feature inconsistency. Moreover, differences in intensity scales and noise profiles across modalities complicate feature fusion. Addressing these issues requires robust image registration algorithms, intensity normalization, and advanced fusion strategies such as attention-based D models. Harmonization techniques and domain adaptation are also critical for mitigating variability in multi-center datasets, ensuring that integrated models generalize effectively across diverse clinical environments.

Fusion models use DL to merge features from different modalities. For example, CNN-based feature fusion models combine PET and MRI data for improved glioblastoma segmentation, while Transformer-based multi-modal networks integrate histopathological and radiological images for breast cancer diagnosis [76].

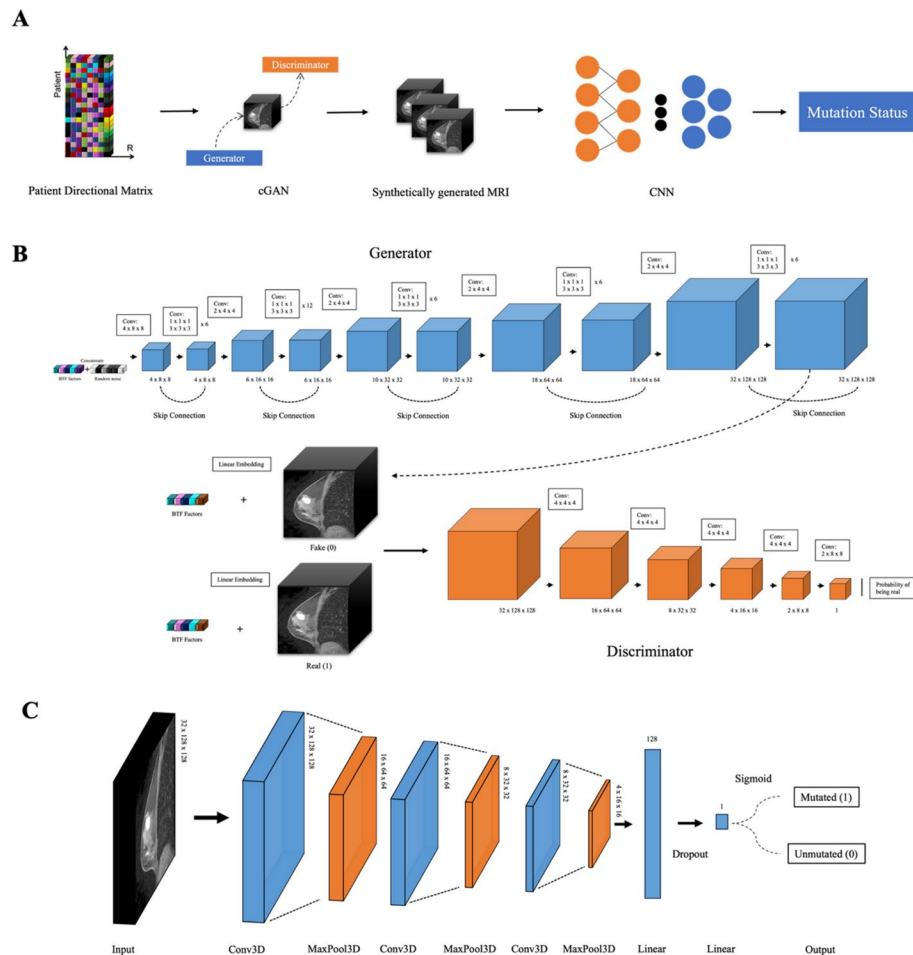


Fig. 4 Architecture diagram of the DL framework for radiogenomic mutation prediction. (A) Overall study design. A patient-specific directional tensor comprising 17 latent features is input into a cGAN to generate synthetic MRI images and support downstream mutation status prediction. (B) Schematic of the cGAN architecture used for MRI synthesis. (C) Workflow for mutation classification: both the labeled synthetic MRI generated by the cGAN and real patient MRI images from The Cancer Imaging Archive (TCIA) are processed through multiple deep learning layers to yield a binary classification output indicating the presence or absence of tumor-driving gene mutations. <https://translational-medicine.biomedcentral.com/articles/https://doi.org/10.1186/s12967-024-05018-9>. Reproduced from reference [73] under the Creative Commons Attribution 4.0 International License (<https://creativecommons.org/licenses/by/4.0/>) (accessed on 12 March 2026)) with no change of the original content

Multi-modal ML models are used to assess metastatic disease and treatment efficacy. AI algorithms analyzing whole-body PET-CT scans detect distant metastases in lung and prostate cancer, while DL models track tumor changes over time in longitudinal imaging studies [77].

3.4 CAD systems

CAD systems use ML to assist radiologists in diagnosing cancer, reducing interpretation variability and improving workflow efficiency. These systems are particularly valuable in screening programs, where large volumes of imaging data must be analyzed. ML-powered CAD systems highlight suspicious lesions, rank possible diagnoses, and provide confidence scores to support clinical decision-making [78]. These systems are used in both 2D and 3D imaging, helping radiologists prioritize high-risk cases for further review [79].

AI-assisted mammography and lung nodule detection have shown significant advancements. For mammography, AI algorithms developed by organizations like Google Health [80] and DeepMind [81] assist in detecting breast cancer by analyzing mammograms with accuracy comparable to expert radiologists. In lung nodule detection, CAD systems for chest CT scans, such as those used in the National Lung Screening Trial (NLST), enhance the detection of early-stage lung cancer by flagging suspicious nodules [82].

Studies have shown that AI-assisted diagnosis improves sensitivity and specificity while reducing false positives and false negatives [83–85]. ML models help mitigate inter-observer variability and enhance efficiency, particularly in high-volume screening settings. However, challenges remain in ensuring interpretability and gaining regulatory approval for widespread clinical adoption.

Comparative evaluations indicate that AI-assisted imaging can complement expert readers, reduce inter-observer variability, and lower error rates in routine practice. In population breast screening, a nationwide deployment reported that radiologists using AI triage detected more cancers with fewer false-positive recalls than standard double reading, suggesting improved specificity without loss of sensitivity [86]. Similarly, in chest radiography, an AI “second reader” identified false-negative examinations initially reported as normal, demonstrating potential to reduce misses in day-to-day workflow [87]. Beyond single-task readers, multi-annotator/ensemble frameworks have shown higher segmentation accuracy than any single annotator, offering a practical path to standardize labels and mitigate inter-reader variation [88]. Professional guidance from major radiology societies further recommends continuous registry-based monitoring of sensitivity/specificity and subgroup performance after deployment, to limit drift and sustain clinical safety, reinforcing that AI is most effective as a calibrated, audited aid to expert judgment rather than a replacement [89].

4 Machine learning in cancer therapy planning and monitoring

ML is transforming cancer therapy by improving treatment planning, monitoring therapeutic response, and guiding real-time interventions. AI-driven models enhance radiotherapy planning, predict treatment outcomes, and optimize image-guided interventions, leading to more personalized and adaptive cancer treatments.

4.1 AI-assisted radiotherapy planning

Radiotherapy is a cornerstone of cancer treatment, requiring precise dose delivery to tumor tissues while sparing healthy organs. ML algorithms automate and optimize RT planning by segmenting anatomical structures, predicting optimal dose distributions, and enabling adaptive treatment strategies. Accurate segmentation of tumor volumes and OARs is crucial for radiotherapy planning. DL models, particularly CNN-based architectures such as U-Net and DeepLab, have been widely used for automatic contouring in MRI and CT scans [90]. These models reduce variability in manual delineation and enhance treatment consistency. Transformer-based segmentation models are also emerging, offering improved long-range spatial dependencies for precise boundary detection [91].

AI models facilitate automated dose distribution prediction and treatment plan optimization. Neural networks and reinforcement learning approaches analyze historical treatment plans to generate dose-volume histograms (DVHs) and optimize

intensity-modulated radiotherapy (IMRT) and volumetric modulated arc therapy (VMAT) plans [92, 93]. These models improve dose conformity and reduce planning time while maintaining high treatment quality. Adaptive radiotherapy (ART) can significantly benefit from real-time AI-driven adjustments based on continuous imaging feedback. By integrating AI models into ART workflows, treatment plans can be dynamically updated in response to anatomical or physiological changes observed during therapy, such as tumor shrinkage or organ motion. For example, daily MRI or CBCT scans can feed into deep learning models that predict dose distribution and automatically re-optimize beam configurations, reducing the need for manual replanning [94]. This closed-loop system enhances treatment precision, minimizes toxicity to surrounding organs, and supports personalized therapy by adapting to patient-specific changes over the course of treatment. Clinically, such an approach could improve local control rates and reduce adverse effects, but it requires robust automation, real-time computation, and rigorous validation to ensure safety and regulatory compliance.

4.2 Predicting treatment response from imaging

Predicting treatment response is essential for guiding therapeutic decisions and personalizing cancer care. ML models extract imaging-based biomarkers to assess tumor response, anticipate treatment-related toxicities, and refine patient-specific therapeutic strategies [95]. Radiomics and DL models analyze pre- and post-treatment imaging to identify predictive biomarkers of therapy response. For example, texture features from MRI and PET scans are correlated with tumor radiosensitivity, while AI-driven feature selection enhances predictive accuracy [96].

AI is playing a growing role in immunotherapy response assessment. ML models analyze radiomics features and genetic data to predict patient response to immune checkpoint inhibitors (ICIs) [97]. DL also assists in personalizing targeted therapies by integrating imaging, genomics, and clinical data to identify optimal treatment strategies [98].

ML models predict radiation-induced toxicities such as pneumonitis, mucositis, and fibrosis. AI-driven dose-response modeling helps identify patients at risk of severe side effects, enabling proactive treatment modifications to minimize toxicity while maintaining therapeutic efficacy [99].

Effective treatment monitoring is not only about assessing response but also about informing timely and adaptive intervention. In clinical practice, imaging-based response assessments, such as MRI performed every 48–72 h, provide critical insights that can trigger changes in therapeutic strategy, such as treatment intensification or replanning of radiotherapy. However, the true clinical value lies in closing the loop: using monitoring data to guide interventions and then recalibrating AI models based on post-intervention imaging. This iterative feedback mechanism allows AI systems to dynamically update confidence scores and predictive outputs based on new anatomical or physiological changes.

RL provides a powerful framework for implementing AI-driven adaptive radiotherapy systems that continuously refine treatment strategies based on imaging feedback and observed outcomes. In this approach, the AI agent interacts with the treatment environment, where each adjustment, such as modifying dose distribution or beam angles, is evaluated through a reward function linked to tumor control and toxicity reduction.

Over time, RL models can learn optimal policies for patient-specific adaptation, creating a closed-loop system that responds dynamically to anatomical and biological changes during therapy. For example, model-free RL could optimize daily plan adjustments using real-time MRI or CBCT data, while model-based RL could simulate future treatment scenarios to guide proactive decisions. However, clinical translation requires addressing challenges such as defining clinically meaningful reward functions, ensuring patient safety during exploration, and integrating RL algorithms into time-sensitive workflows [43].

To bridge monitoring and intervention, AI-driven workflows must operate within an integrated framework where continuous imaging feeds decision-making engines, and model outputs guide precision interventions. These actions, in turn, are evaluated by subsequent imaging, creating a closed-loop, adaptive cancer therapy cycle. Figure 5 shows such a closed-loop workflow illustrating how treatment monitoring informs image-guided interventions, followed by re-monitoring and model recalibration for adaptive cancer therapy.

4.3 Image-guided interventions

Real-time imaging is essential for guiding surgical and radiotherapy interventions. AI-powered systems enhance intraoperative imaging, robotic-assisted surgery, and motion management, improving treatment accuracy and safety. AI-driven imaging enhances intraoperative decision-making by providing real-time tumor localization and margin assessment [100]. DL models integrate US, MRI, and optical imaging to guide resection in breast and brain tumor surgeries [101]. In radiotherapy, AI-assisted fluoroscopy and cone-beam CT (CBCT) improve tumor tracking for precise dose delivery [102].

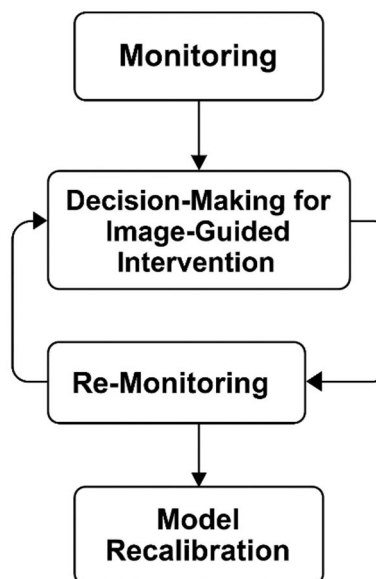


Fig. 5 Flowchart illustrating the closed-loop process of AI-assisted cancer treatment. The loop integrates periodic monitoring (e.g., MRI every 72 h), decision-making for image-guided intervention, and re-monitoring to evaluate intervention outcomes. This continuous feedback cycle enables model recalibration, improved treatment precision, and adaptive response strategies

ML enhances robotic-assisted surgery by improving surgical planning and intraoperative guidance. AI-based motion tracking and real-time anatomical mapping refine robotic procedures such as prostatectomy, ensuring precise tumor resection while minimizing damage to surrounding tissues [103].

Patient motion during radiotherapy affects treatment accuracy. AI-based motion prediction models analyze respiratory patterns and organ deformation to guide real-time dose adjustments [104]. DL frameworks, such as RNNs and Transformer models, predict motion trajectories, improving tumor targeting in lung and abdominal cancers [105].

5 Challenges and limitations

Despite the significant advancements in ML for cancer imaging and therapy, several challenges remain in ensuring the robustness, fairness, and clinical adoption of AI-driven tools. These challenges include data scarcity, model generalization issues, and regulatory hurdles. Addressing these limitations is crucial for the successful integration of ML into routine clinical practice.

5.1 Data scarcity and annotation challenges

ML models require large, diverse, and well-annotated datasets for effective training and validation [106]. However, acquiring such datasets in medical imaging remains a major challenge due to data privacy concerns, limited publicly available datasets, and the labor-intensive nature of manual annotation.

Medical imaging datasets often suffer from limited sample sizes, especially for rare cancers. Unlike natural image datasets, where millions of labeled images are available, medical datasets are restricted due to patient confidentiality and institutional policies [107]. Furthermore, expert annotations, such as tumor delineations and radiological assessments, require significant time and domain expertise, leading to inconsistent labeling across different datasets [108].

To address data scarcity, researchers have adopted several strategies. GANs create synthetic medical images to augment existing datasets, enhancing model robustness. Beyond general data augmentation, synthetic data generation using GANs has shown particular promise in addressing the scarcity of imaging data for rare cancer subtypes. GAN-based frameworks can generate high-fidelity synthetic images that mimic the anatomical and pathological characteristics of underrepresented tumors, thereby balancing class distributions and reducing model bias. For example, cGANs can be trained to produce modality-specific images (e.g., MRI or CT) conditioned on tumor type or clinical attributes, enabling targeted augmentation for rare cancers such as sarcomas or pediatric tumors. These synthetic datasets not only expand training diversity but also improve model generalization across heterogeneous clinical settings. For example, synthetic MRI and CT scans have been utilized to improve training data for tumor segmentation tasks [109]. Transfer learning involves fine-tuning pre-trained DL models, initially trained on large-scale image datasets like ImageNet [110], on smaller medical datasets, which boosts performance even with limited labeled data. Federated learning, on the other hand, enables AI models to be trained across multiple institutions without centralizing medical data, thus preserving patient privacy and improving model generalization [111].

5.2 Model generalization and bias issues

ML models must generalize well across different patient populations, imaging protocols, and institutions. However, AI models often struggle with distribution shifts, leading to reduced performance in real-world clinical settings.

Variability in medical imaging arises from differences in scanner manufacturers, imaging protocols, and demographic factors. ML models trained on data from a single institution may not perform reliably when applied to external datasets [112]. Domain adaptation techniques, such as adversarial training and style transfer, are being explored to enhance model robustness across diverse clinical settings [113].

Bias in AI models can stem from imbalanced training data, resulting in uneven performance across various patient groups. For instance, research indicates that AI-based mammography screening models may not perform as well for patients from underrepresented ethnic backgrounds [114]. To promote fairness, researchers should use diverse and representative datasets during model training, implement bias mitigation techniques like reweighting strategies and fairness-aware algorithms, and conduct external validation studies to evaluate model performance across different demographic groups and clinical settings [115].

5.3 Regulatory and clinical adoption hurdles

For ML-based medical imaging tools to be deployed in clinical practice, they must meet regulatory standards and address ethical considerations. The pathway to regulatory approval is complex, requiring rigorous validation, explainability, and risk assessment.

Regulatory bodies like the U.S. Food and Drug Administration (FDA) and the European Medicines Agency (EMA) have set guidelines for AI-based medical devices [116]. Key approval requirements include demonstrating clinical safety and efficacy through large-scale validation studies, ensuring transparency in model decision-making for interpretability and explainability, and continuously monitoring AI systems post-deployment to assess real-world performance [117]. Examples of AI-powered imaging tools that have received FDA clearance include AI-based mammography screening systems [118] and DL models for detecting strokes in CT scans [119]. Figure 6 shows all the FDA approved AI/machine learning-based devices and technologies [117].

The integration of AI into medical decision-making brings up ethical concerns related to patient autonomy, liability, and data privacy [120, 121]. Key challenges include determining accountability in cases where AI-assisted diagnoses result in misdiagnosis or treatment errors, ensuring transparency so that AI decisions are interpretable by clinicians to build trust in automated systems, and adhering to data privacy regulations like the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA) to protect patient information [122]. While AI has the potential to improve cancer imaging and therapy, addressing these regulatory and ethical challenges is crucial for responsible AI deployment in clinical practice.

5.4 Barriers to clinical translation and routine deployment

Despite rapid methodological advances and promising retrospective results, the clinical adoption of machine learning-based cancer imaging tools remains limited. One of the primary barriers is the gap between algorithm development environments and

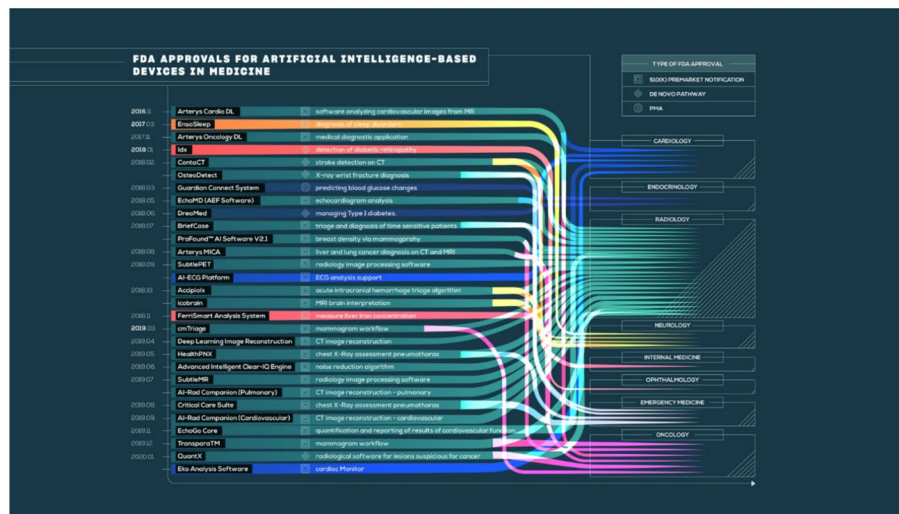


Fig. 6 FDA approvals of AI-based medical devices across specialties and use cases. This infographic summarizes FDA-cleared AI/ML-enabled medical devices from 2016 onward, categorized by application (e.g., diagnosis, image reconstruction), clinical domain (e.g., radiology, cardiology), and approval type (510(k), De Novo, PMA). Each line traces the regulatory and clinical pathway from task to specialty, revealing trends such as the dominance of radiology applications and the growing diversity of clinical domains. This figure underscores the regulatory momentum and real-world deployment potential for AI technologies in healthcare. <https://www.nature.com/articles/s41746-020-00324-0>. Reproduced from reference [117] under the Creative Commons Attribution 4.0 International License (<https://creativecommons.org/licenses/by/4.0/>) (accessed on 12 March 2026)) with no change of the original content

real-world clinical workflows. Most ML models are trained and evaluated on curated, retrospective datasets under controlled conditions, whereas clinical environments are dynamic, heterogeneous, and time-constrained. Differences in scanner vendors, imaging protocols, patient positioning, and institutional practices can lead to substantial performance degradation when models are deployed outside their original training domain [123].

Workflow integration represents another major obstacle [124]. Clinical imaging and treatment planning systems, such as PACS, radiotherapy planning platforms, and electronic health records, operate under strict interoperability, latency, and safety requirements. ML tools that require manual data transfer, offline processing, or extensive human intervention disrupt established workflows and are unlikely to be adopted, regardless of algorithmic performance. Real-time or near-real-time inference, particularly in adaptive radiotherapy or image-guided interventions, further amplifies these constraints.

Trust and interpretability also remain critical limiting factors [125]. Many deep learning models function as “black boxes,” providing predictions or segmentations without transparent reasoning. In high-stakes oncology decision-making, clinicians must be able to understand, verify, and contextualize AI outputs. Without robust explainability, uncertainty quantification, and calibration, clinicians may be reluctant to rely on AI-assisted recommendations, particularly when model outputs conflict with clinical judgment.

Regulatory and liability considerations further complicate clinical deployment [126]. Regulatory approval requires extensive validation, often including multi-center and prospective studies, which are resource-intensive and time-consuming. Moreover,

responsibility and accountability remain unclear when AI-assisted decisions contribute to diagnostic or treatment errors. These concerns slow institutional adoption and necessitate human-in-the-loop frameworks rather than fully autonomous systems.

Finally, the lack of prospective and adaptive validation limits clinical confidence [127]. Many published ML models are evaluated using retrospective datasets and static endpoints, whereas clinical care requires continuous monitoring, feedback, and model updating. Without closed-loop validation and post-deployment performance monitoring, AI systems risk model drift and reduced safety over time. Addressing these challenges will require not only technical innovation, but also workflow-aware design, clinician engagement, regulatory alignment, and prospective clinical evaluation.

Table 2 summarizes the key challenges in ML for cancer imaging and therapy, encompassing data scarcity, model generalization and bias, regulatory constraints, and barriers to clinical translation and routine deployment. While methodological approaches such as generative data augmentation, transfer learning, domain adaptation, and compliance with FDA and GDPR requirements can enhance technical robustness, successful clinical adoption requires addressing broader system-level constraints. In particular, translation into routine practice depends on reliable generalization across institutions, prospective and adaptive validation, and seamless interoperability with clinical infrastructure such as PACS and radiotherapy planning systems. Moreover, effective integration into clinical workflows hinges on clinician trust, interpretability, uncertainty awareness, and human-in-the-loop decision support rather than fully autonomous operation. FDA-cleared AI tools for stroke detection and mammography triage [118, 119] illustrate that adherence to safety standards, explainability, and real-world workflow compatibility are essential for sustained clinical acceptance.

Table 2 Summary of challenges and limitations in machine learning for cancer imaging and therapy

Challenge category	Key issues	Clinical/technical implications	Representative mitigation strategies	Key references
Data scarcity and annotation	Limited dataset size; scarcity of rare cancer cases; high cost and variability of expert annotations	Overfitting, reduced robustness, poor performance on rare cancers	Synthetic data generation using GANs; transfer learning from large-scale datasets; federated learning for multi-institutional training	[106–111]
Model generalization and bias	Distribution shifts across scanners, protocols, and populations; demographic and institutional bias	Performance degradation in external cohorts; unequal accuracy across patient subgroups	Domain adaptation (e.g., adversarial training, style transfer); diverse datasets; fairness-aware learning; external validation	[112–115]
Regulatory and ethical constraints	Complex approval pathways; explainability requirements; data privacy and accountability concerns	Delayed deployment; limited trust; restricted autonomy of AI systems	Compliance with FDA/EMA guidance; explainable AI; post-market surveillance; adherence to GDPR and HIPAA	[116–122]
Clinical translation and routine deployment barriers	Workflow incompatibility with PACS/EHRs; lack of real-time capability; black-box behavior; unclear liability; absence of prospective and adaptive validation	Low adoption despite high retrospective performance; clinician distrust; safety risks due to model drift	Workflow-aware design; real-time and interoperable systems; explainable and uncertainty-aware AI; human-in-the-loop frameworks; prospective and continuous validation	[123–127]

6 Future perspectives and emerging trends

The field of ML in cancer imaging is evolving rapidly, with new advancements aiming to enhance the interpretability, privacy, and computational efficiency of AI-driven models. Future research directions focus on XAI, federated learning for privacy-preserving AI, and the potential integration of quantum computing for advanced image processing. These emerging trends aim to address current limitations and push the boundaries of AI applications in oncology.

6.1 XAI in cancer imaging

As AI models become increasingly integrated into clinical workflows, ensuring their interpretability is essential for trust and adoption by healthcare professionals. XAI techniques aim to make DL models more transparent, allowing radiologists and oncologists to understand the reasoning behind AI-generated predictions [128].

One of the primary concerns in deploying AI in cancer imaging is the “black box” nature of DL models. Without clear explanations for AI-based diagnoses or segmentations, clinicians may be reluctant to fully rely on these models. Explainable AI methods provide visual or quantitative insights into model decisions, ensuring that AI-generated recommendations align with radiologists’ clinical expertise.

In clinical settings, XAI tools such as Grad-CAM and SHAP have demonstrated practical utility in assisting radiologists with decision-making. Grad-CAM generates heatmaps that highlight the regions of medical images most influential in a model’s prediction, enabling clinicians to visually verify whether the AI system is focusing on relevant tumor areas. For instance, in breast MRI or lung CT scans, Grad-CAM can emphasize suspicious lesions, helping radiologists corroborate AI-generated diagnoses with anatomical evidence [129]. SHAP, on the other hand, provides feature-level explanations by quantifying the contribution of each radiomic feature to the model’s output. This is particularly useful in radiomics-based classification tasks, where SHAP values can reveal which texture or intensity features are driving predictions of malignancy or treatment response [130]. By integrating these tools into diagnostic workflows, clinicians gain transparency into AI decisions, fostering trust and enabling more informed, collaborative decision-making [131].

For AI models to be widely accepted in clinical practice, they must align with human decision-making processes. Trust in AI-assisted cancer diagnosis and treatment planning can be fostered by integrating XAI techniques, allowing radiologists and oncologists to validate AI predictions against their own expertise. Future research should focus on refining these methods to enhance their usability and reliability in real-world applications [132].

Clinical deployment of ML models in oncology hinges not only on predictive accuracy, but also on model explainability, the ability to justify or trace decisions to specific inputs. For example, Grad-CAM enables heatmap visualizations that highlight the tumor region influencing a classification, while SHAP scores offer insight into the contribution of each radiomic feature to a prediction [129, 130]. Such tools help radiologists and radiation oncologists validate AI recommendations against their own clinical reasoning, thereby fostering trust and interpretability.

However, explainability alone is insufficient without trustworthiness. Models must be externally validated on diverse populations, calibrated to output reliable probability

estimates, and able to quantify uncertainty when confidence is low. Incorporating these dimensions ensures that AI tools remain transparent, fair, and clinically dependable, particularly in high-risk decision-making scenarios.

6.2 Federated learning for privacy-preserving AI in imaging

Data privacy remains a significant challenge in medical imaging research, as patient data is highly sensitive and subject to strict regulations. Federated learning offers a novel approach to AI model training that allows multiple institutions to collaborate without sharing raw patient data, thereby preserving privacy while improving model generalization.

Traditional AI training requires centralized datasets, which can be challenging to obtain due to ethical and legal restrictions [133]. Federated learning enables AI models to be trained across multiple hospitals or research centers without data leaving local servers. Instead of sharing raw images, only model updates (e.g., gradient changes) are exchanged, ensuring patient data remains secure.

Federated learning has the potential to accelerate AI development in cancer imaging by using diverse datasets from various institutions [134]. Federated learning must also contend with the challenge of heterogeneous data across institutions, including differences in scanner types, imaging protocols, and patient demographics. These variations can lead to non-IID (independent and identically distributed) data, which may degrade model performance and convergence. To address this, domain adaptation techniques such as adversarial training and feature alignment are employed to harmonize feature distributions across sites. Moreover, data harmonization strategies, including intensity normalization and protocol standardization, help reduce inter-institutional variability. Personalized federated learning approaches, which tailor model updates to local data characteristics, are also being explored to improve performance in diverse clinical environments. These methods collectively enhance the robustness and generalizability of federated models, making them more suitable for real-world deployment in multi-center cancer imaging studies. This approach enhances model robustness by incorporating variations in imaging protocols, scanner types, and patient demographics. Applications include AI-assisted tumor detection models trained across multiple hospitals while maintaining patient privacy [135], federated radiomics studies to improve cancer classification and treatment response prediction [136], and collaborative AI research networks to develop generalizable models for rare cancer subtypes [137].

Despite its promise, federated learning faces several challenges. Data heterogeneity, stemming from variations in imaging protocols and scanner quality across institutions, can introduce biases that affect model performance, necessitating standardization methods to harmonize data distributions [138]. In addition, communication efficiency is a concern, as federated learning requires frequent model updates across multiple nodes, which can be computationally expensive and demand high-bandwidth networks. Optimized communication protocols are essential to reduce latency and improve training efficiency.

6.3 Integration with quantum computing for advanced image processing

Quantum computing has the potential to revolutionize cancer imaging by accelerating complex computations in medical image processing, pattern recognition, and

optimization tasks. While still in its early stages, quantum machine learning (QML) is being explored for applications in radiology and oncology [139].

Quantum computing offers exponential speedup for certain computations compared to classical computing. In cancer imaging, QML could enhance image reconstruction by accelerating MRI and PET image processing, thereby reducing scan times while maintaining high-resolution imaging. Quantum Neural Networks (QNNs) can accelerate tasks like MRI or PET image reconstruction by leveraging quantum parallelism, reducing computation time while preserving image quality. In radiomics, QNNs may enable more efficient extraction of complex imaging biomarkers from high-dimensional data. However, practical adoption faces major hurdles, including limited quantum hardware, high error rates, and the need for hybrid quantum-classical frameworks compatible with clinical workflows. In addition, regulatory compliance and standardized benchmarks are essential before clinical deployment. Quantum-enhanced algorithms could also improve radiomics feature extraction, identifying subtle imaging biomarkers for precision oncology [140]. Furthermore, quantum computing could optimize radiotherapy dose distribution calculations, leading to more personalized treatment plans [141].

Current research is investigating the integration of quantum computing with DL for various cancer imaging applications. Quantum-enhanced pattern recognition, utilizing quantum support vector machines (QSVMs) and quantum neural networks QNNs [142], may improve cancer classification from histopathology and radiology images. Quantum-inspired generative models, such as Quantum Boltzmann machines and quantum GANs, could generate high-quality synthetic medical images for data augmentation in ML [143]. Moreover, quantum optimization in imaging pipelines, through variational quantum algorithms, can optimize imaging workflows and reduce computational costs in large-scale medical image processing [144].

Although quantum computing in medical imaging is still in its early stages, future research should prioritize developing quantum-compatible imaging datasets for benchmarking QML algorithms, collaborating with quantum hardware developers to create specialized algorithms for medical image processing [145], and investigating hybrid quantum-classical AI models to leverage the strengths of both computational paradigms [146].

As ML continues to transform cancer imaging, future developments in explainable AI, privacy-preserving federated learning, and quantum computing will further enhance the accuracy, efficiency, and trustworthiness of AI-driven models. Overcoming current limitations and advancing these emerging trends will be key to ensuring that AI-driven innovations in cancer diagnosis and therapy planning reach their full clinical potential.

Compared to recent state-of-the-art reviews [18–20], which primarily emphasize algorithmic advancement and diagnostic performance benchmarks, our review extends the discussion into practical workflow engineering, such as AI-assisted treatment monitoring, real-time interventions, and model recalibration strategies. This system-level view aims to complement existing literature and inform both AI researchers and clinical practitioners on actionable deployment pathways.

For AI to become a routine component of clinical oncology, future research must prioritize not only model accuracy but translational design, including interoperability, clinical workflow integration, and post-deployment monitoring. AI models that are

explainable, continuously learning, and embedded within multidisciplinary care processes will likely have the greatest real-world impact.

Looking ahead, several technically grounded research directions hold promise for advancing the role of ML in cancer imaging and therapy. One key priority is the integration of uncertainty quantification into predictive models. By incorporating Bayesian methods or confidence-aware loss functions, future AI systems can express calibrated probabilities rather than binary predictions, enabling clinicians to interpret model outputs with appropriate caution in high-stakes scenarios. Another direction involves the development of self-supervised and few-shot learning techniques tailored for rare cancer types, where annotated datasets are scarce. These approaches, including contrastive learning and generative pretraining, could improve model performance in underrepresented clinical contexts. Federated learning remains a promising solution for privacy-preserving collaboration across institutions; however, its effectiveness will depend on the incorporation of domain adaptation techniques to manage variability in imaging protocols, scanner types, and demographics. Moreover, the integration of quantum-enhanced learning, such as quantum neural networks or quantum-augmented radiomics, offers a frontier for accelerating imaging analysis and reconstruction tasks, though this remains largely experimental. Finally, clinical translation will depend not only on algorithmic performance but also on the development of regulatory-grade pipelines that support reproducibility, transparency, post-deployment monitoring, and interoperability with existing clinical systems. Addressing these areas will be essential for the safe, effective, and scalable adoption of AI in oncology.

6.4 Open research gaps and future directions

Despite rapid progress in ML methodologies for cancer imaging, several critical research gaps remain that limit clinical translation and long-term impact [147]. One major gap is the scarcity of prospective and adaptive validation studies. Most ML models are developed and evaluated retrospectively using static datasets, whereas clinical oncology requires models that can operate reliably in dynamic environments, adapt to anatomical and biological changes over time, and maintain performance under distribution shifts.

A second key gap is the limited integration of uncertainty quantification and confidence-aware decision support [148]. Current models often produce point estimates without communicating uncertainty, which is problematic in high-stakes clinical settings where risk-aware interpretation is essential. Future research should prioritize probabilistic modeling, calibration strategies, and uncertainty-aware outputs that align with clinical decision-making processes.

Workflow-aware model design remains another underexplored area [149]. Many ML systems are developed independently of clinical constraints such as time sensitivity, interoperability with imaging and treatment planning systems, and human-in-the-loop requirements. Research that explicitly incorporates workflow integration, real-time inference, and clinician interaction into model design is essential for meaningful deployment.

Standardization and benchmarking also represent persistent gaps [150]. Variability in datasets, preprocessing pipelines, evaluation metrics, and reporting practices makes it difficult to compare models and assess reproducibility across studies. Community-driven

Table 3 Future perspectives, emerging trends, and open research gaps in machine learning for cancer imaging

Theme	Core focus	Key opportunities	Outstanding challenges/gaps	Key references
Explainable AI (XAI)	Improving transparency and interpretability of ML models for clinical trust	Visual (Grad-CAM) and feature-level (SHAP) explanations; alignment with clinician reasoning; improved human–AI collaboration	Limited external validation; lack of uncertainty quantification; usability and reliability in real-world clinical workflows	[128–132]
Federated learning for privacy-preserving AI	Multi-institutional model training without sharing raw patient data	Privacy preservation; improved generalizability; collaborative learning for rare cancers	Non-IID data across sites; scanner/protocol heterogeneity; communication and scalability constraints	[133–138]
Quantum computing and quantum ML	Accelerating computation for image reconstruction, radiomics, and optimization	Faster MRI/PET reconstruction; advanced radiomics; optimization of radiotherapy planning	Immature hardware; high error rates; lack of benchmarks; need for hybrid quantum–classical pipelines	[139–146]
Open research gaps and future directions	Translational bottlenecks limiting real-world deployment	Prospective and adaptive validation; uncertainty-aware decision support; workflow-aware model design; standardized benchmarking; post-deployment monitoring	Predominance of retrospective studies; benchmarking crisis; limited lifecycle governance; insufficient regulatory-grade pipelines	[147–151]

benchmarks, standardized validation protocols, and multi-institutional evaluation frameworks are needed to support robust comparison and regulatory readiness.

Finally, post-deployment monitoring and model lifecycle management remain insufficiently studied [151]. Few works address how ML models should be monitored, updated, or recalibrated after clinical deployment to mitigate performance drift and maintain safety. Future research should focus on continuous learning frameworks, audit mechanisms, and governance models that support the safe evolution of AI systems in clinical practice. Addressing these research gaps will be critical for transitioning machine learning in cancer imaging from proof-of-concept studies to reliable, adaptive, and clinically integrated tools. Table 3 summarizes the major emerging trends and forward-looking research directions in ML for cancer imaging, encompassing explainable AI for model transparency, federated learning for privacy-preserving collaboration, quantum computing for advanced image processing, and critical open research gaps that shape future development. Together, these areas highlight both technological opportunities and unresolved challenges, including the need for uncertainty-aware modeling, workflow-integrated system design, standardized benchmarking, and post-deployment monitoring. Addressing these issues will be essential for enhancing model interpretability, data security, computational efficiency, and translational readiness, thereby supporting the safe and scalable integration of AI into clinical oncology.

7 Conclusion

The motivation for this review stems from the growing need to harness ML and AI to address critical challenges in cancer imaging, including early detection, accurate segmentation, and personalized therapy planning. As cancer remains a leading cause of mortality worldwide, improving imaging-based diagnosis and treatment through advanced computational methods is essential for enhancing patient outcomes.

This review has provided a comprehensive overview of ML principles in medical imaging, highlighting their transformative impact on tumor detection, segmentation, radiomics, multi-modal imaging, and therapy planning. Over the past decade, AI-driven approaches, such as CNNs, Transformer-based architectures, and GANs, have demonstrated remarkable potential in improving diagnostic accuracy, automating complex imaging tasks, and enabling personalized treatment strategies. Furthermore, AI-powered radiomics and multi-modal integration have facilitated the extraction of clinically meaningful biomarkers, supporting predictive modeling for treatment response.

Despite these advancements, significant challenges remain before AI can be fully integrated into routine clinical workflows. Issues such as data scarcity, annotation complexity, generalization across diverse populations, and ethical and regulatory considerations must be addressed to ensure safe and effective deployment. Future research should prioritize XAI techniques to enhance interpretability and clinician trust, privacy-preserving methods like federated learning for secure multi-institutional collaboration, and emerging technologies such as quantum computing to accelerate complex imaging computations.

The successful translation of ML in cancer imaging from research to clinical practice will require interdisciplinary collaboration among AI researchers, radiologists, radiation oncologists, and regulatory bodies. By addressing current limitations and leveraging innovative approaches, AI-driven solutions have the potential to revolutionize cancer imaging, enabling earlier detection, more accurate diagnosis, and personalized treatment planning, ultimately improving patient care and outcomes in oncology. Beyond methodological innovation, future progress will depend on addressing key research gaps related to prospective validation, uncertainty-aware modeling, workflow integration, and post-deployment governance. Bridging these gaps is essential for ensuring that AI-driven imaging tools achieve sustained clinical impact rather than remaining confined to retrospective research settings.

Abbreviations

AI	Artificial intelligence
CAD	Computer-aided diagnosis
CNN	Convolutional neural network
CT	Computed Tomography
DL	Deep learning
DSC	Dice similarity coefficient
DVH	Dose-Volume Histogram
EMA	European Medicines Agency
FDA	Food and Drug Administration
GAN	Generative Adversarial Network
GDPR	General Data Protection Regulation
HIPAA	Health Insurance Portability and Accountability Act
IMRT	Intensity-Modulated Radiotherapy
IoU	Intersection over Union
LIME	Local Interpretable Model-Agnostic Explanations
LSTM	Long Short-Term Memory
ML	Machine learning
MRI	Magnetic Resonance Imaging
OAR	Organ at risk
PET	Positron Emission Tomography
RNN	Recurrent Neural Network
RT	Radiotherapy
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
TCIA	The Cancer Imaging Archive
US	Ultrasound
VMAT	Volumetric Modulated Arc Therapy
VIT	Vision Transformer

XAI Explainable Artificial Intelligence
mpMRI Multiparametric Magnetic Resonance Imaging

Author contributions

Conceptualization, J.C.L.C.; methodology, J.C.L.C.; validation, J.C.L.C.; formal analysis, J.C.L.C.; investigation, J.C.L.C.; resources, J.C.L.C.; data curation, J.C.L.C.; writing—original draft preparation, J.C.L.C.; writing—review and editing, J.C.L.C.; visualization, J.C.L.C.; supervision, J.C.L.C.; project administration, J.C.L.C.; funding acquisition, J.C.L.C. All authors have read and agreed to the published version of the manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

This article is a narrative review and does not involve human participants, animals, or patient data. Ethics approval was not required. This study does not include human participants. Consent to participate is not applicable.

Consent for publication

This review contains no individual person's data or identifiable images. Consent for publication is not applicable.

Competing interests

The authors declare no competing interests.

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